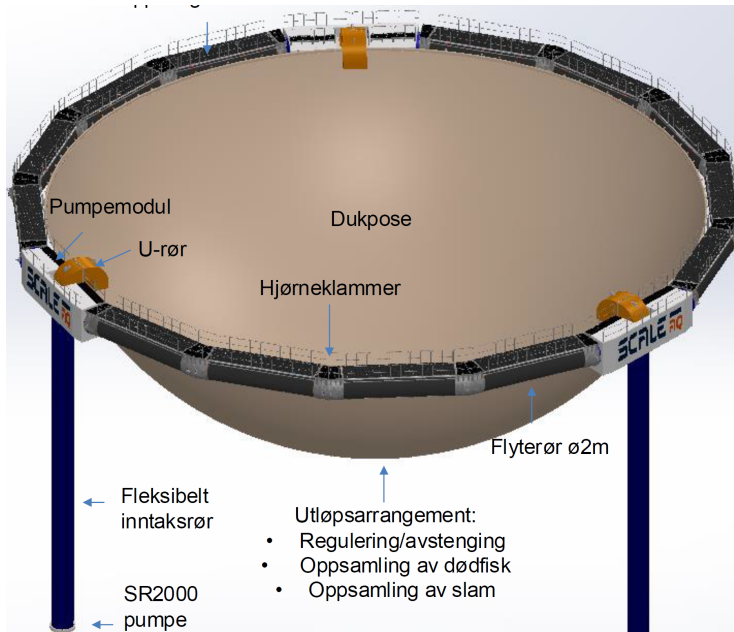


Design specification

Closed Aquaculture System

Operational systems - ScaleAQ

May 20, 2025



Functional requirements

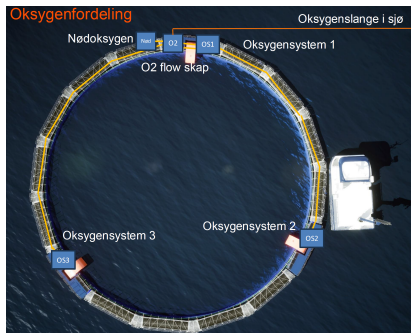


FUNCTIONAL REQUIREMENT SPECIFICATION FOR PUMP

INLET SYSTEM: Level control

1. Process: Regulate the internal water level relative to the outer sea level (delta height). Required to accurately measure levels down to 1mm.
2. Inputs:
Field instruments:
Pressure, distance (ultrasonic), RPM, Power, valve position
User:
delta height set point, RPM, regulator parameters, control commands
3. Outputs: RPM control signal to pumps, control commands to FRAMO PLC controller
4. Major error: Loss of IO inputs, fatal error on VSD's, disturbance due to weather/waves
5. Exception response: Switch to manual operation with default values, trip VSD (2oo3)

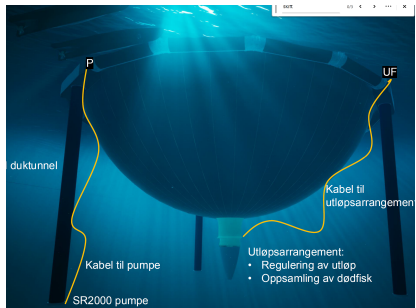
Functional requirements



FUNCTIONAL REQUIREMENT SPECIFICATION FOR OXYGENATOR INJECTOR

1. Process: Inject oxygen into branch of each inlet pump.
2. Inputs:
Field instruments:
 O_2 concentration, pressure,
User:
oxygen concentration set point, regulator parameters, oxygen control valve position
3. Outputs: RPM control signal to pumps, control commands to FRAMO PLC controller
4. Major error: Loss of oxygen reading, loss of pump speed, control valve / solenoid malfunction, centrifugal pump/VSD error
5. Exception response: Switch to manual operation with default values, trip oxygenator system

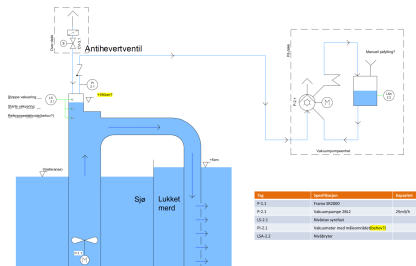
Functional requirements



FUNCTIONAL REQUIREMENT SPECIFICATION FOR OUTLET VALVE

- | | | |
|----|--------------------|---|
| 1. | Process: | Control the water exchange rate |
| 2. | Inputs: | <i>Field instruments:</i>
Outlet valve position, RPM of inlet pumps, power consumptin inlet pumps
<i>User:</i>
set point position outlet valve |
| 3. | Outputs: | Control signal to outlet valve |
| 4. | Major error: | Loss of inlet pumps, outlet valve malfunction, clogging |
| 5. | Exception response | Close outlet valve, trip inlet pumps, reduce water level height on inlet pump system |

Scenarios

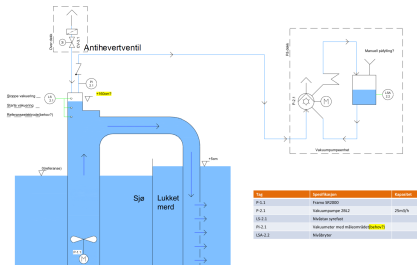


1. All systems are powered up and outlet valve is in position within valid range.

Finish

Scenario 1 - Cold start of inlet pump no. 1 into manual operation mode at a given speed.

Scenarios

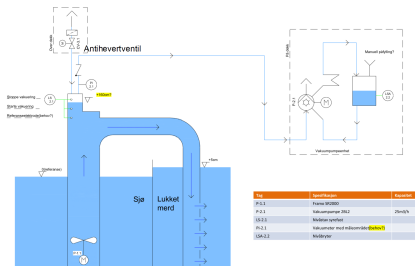


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1

Finish

Scenario 1 - Cold start of inlet pump no. 1 into manual operation mode at a given speed.

Scenarios

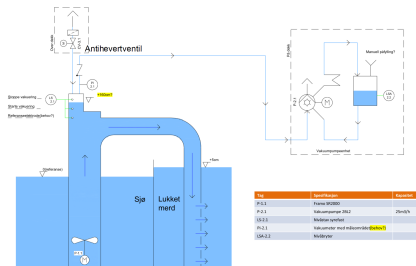


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.

Finish

Scenario 1 - Cold start of inlet pump no. 1 into manual operation mode at a given speed.

Scenarios

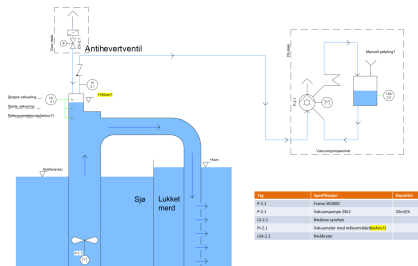


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.
4. The pumps starts at reduced speed, vacuum system starts and anti-siphon valve closes.

Finish

Scenario 1 - Cold start of inlet pump no. 1 into manual operation mode at a given speed.

Scenarios

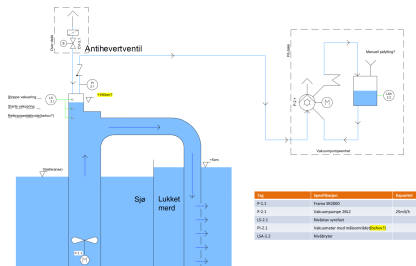


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.
4. The pumps starts at reduced speed, vacuum system starts and anti-siphon valve closes.
5. After a limit switch at outlet piping triggers, the vacuum pump stops and the pump starts running at reduced speed

Finish

Scenario 1 - Cold start of inlet pump no. 1 into manual operation mode at a given speed.

Scenarios

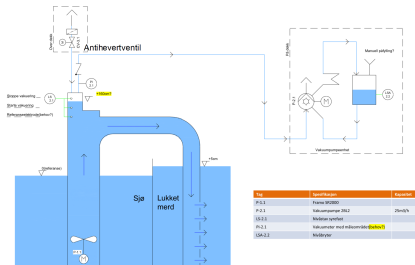


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.
4. The pumps starts at reduced speed, vacuum system starts and anti-siphon valve closes.
5. After a limit switch at outlet piping triggers, the vacuum pump stops and the pump starts running at reduced speed
6. System changes to "remote operation" and in "manual mode" at 50 RPM.

Finish

Scenario 1 - Cold start of inlet pump no. 1 into manual operation mode at a given speed.

Scenarios

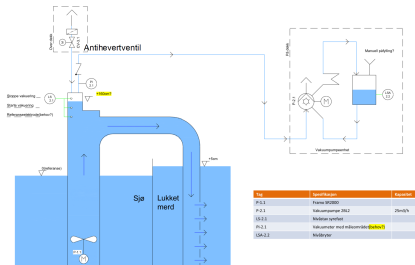


1. All systems are powered up and outlet valve is in position within valid range.

Finish

Scenario 2 - Cold start of inlet pump no. 1 into failed to start state.

Scenarios

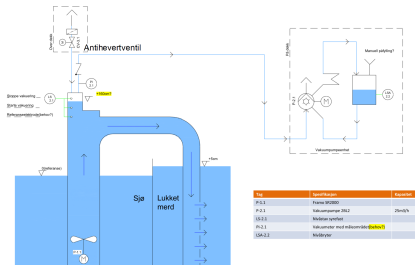


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1

Finish

Scenario 2 - Cold start of inlet pump no. 1 into failed to start state.

Scenarios

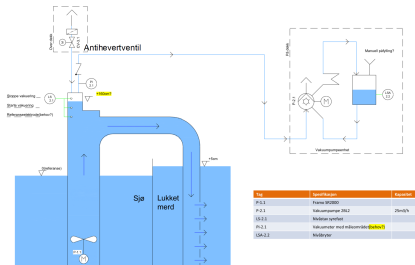


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.

Finish

Scenario 2 - Cold start of inlet pump no. 1 into failed to start state.

Scenarios

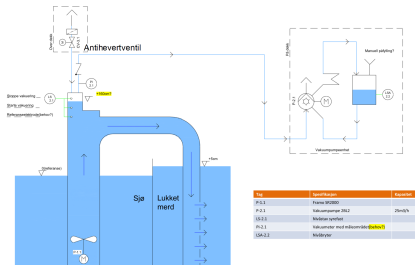


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.
4. The pumps starts at reduced speed, vacuum system starts and anti-siphon valve closes.

Finish

Scenario 2 - Cold start of inlet pump no. 1 into failed to start state.

Scenarios

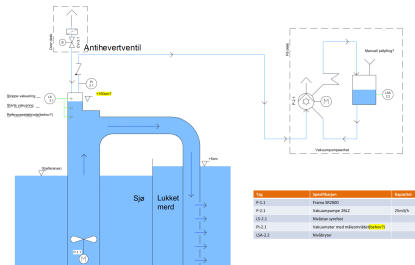


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.
4. The pumps starts at reduced speed, vacuum system starts and anti-siphon valve closes.
5. Limit switch has not triggered within the defined time.

Finish

Scenario 2 - Cold start of inlet pump no. 1 into failed to start state.

Scenarios



1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.
4. The pumps starts at reduced speed, vacuum system starts and anti-siphon valve closes.
5. Limit switch has not triggered within the defined time.
6. System changes to "failed to start", stop command is sent to the pump and anti siphon valve opens.

Finish

Scenario 2 - Cold start of inlet pump no. 1 into failed to start state.

Scenarios

1. All pumps have started and is running in remote operation at 50RPM.

Finish

Scenario 3 - Putting the pumps in auto mode after startup and update the set point to the regulator.

Scenarios

1. All pumps have started and is running in remote operation at 50RPM.
2. Send auto command to controller. Verify that the controller set point is updated with the current value.

Finish

Scenario 3 - Putting the pumps in auto mode after startup and update the set point to the regulator.

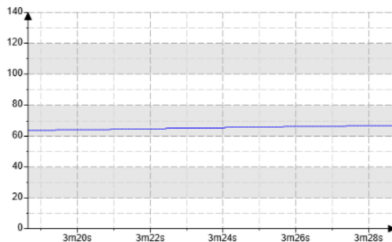
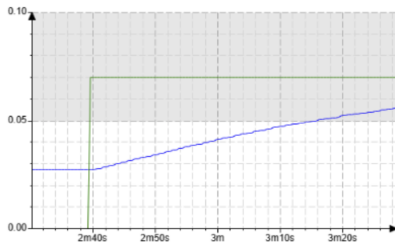
Scenarios

1. All pumps have started and is running in remote operation at 50RPM.
2. Send auto command to controller. Verify that the controller set point is updated with the current value.
3. Verify that the RPM control signal to the pumps are stable at around 50RPM.

Finish

Scenario 3 - Putting the pumps in auto mode after startup and update the set point to the regulator.

Scenarios

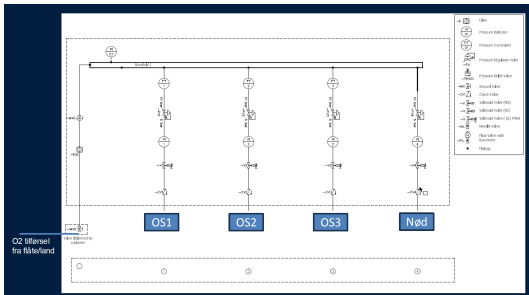


1. All pumps have started and is running in remote operation at 50RPM.
2. Send auto command to controller. Verify that the controller set point is updated with the current value.
3. Verify that the RPM control signal to the pumps are stable at around 50RPM.
4. Change set point to 7cm. Verify that RPM control signal and delta height is stable and rising towards stationary values.

Finish

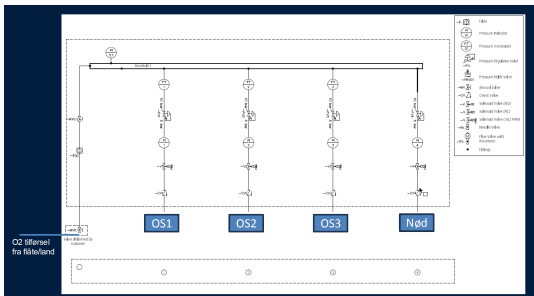
Scenario 3 - Putting the pumps in auto mode after startup and update the set point to the regulator.

Scenarios



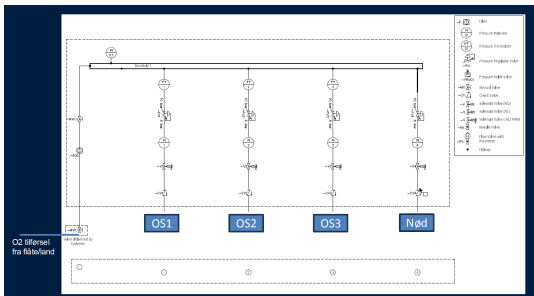
1. Inlet pump no. 1 is running at above minimum speed.

Scenarios



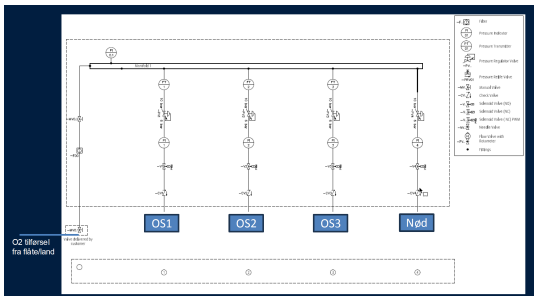
1. Inlet pump no. 1 is running at above minimum speed.
2. We have minimum pressure at oxygen piping manifold.

Scenarios



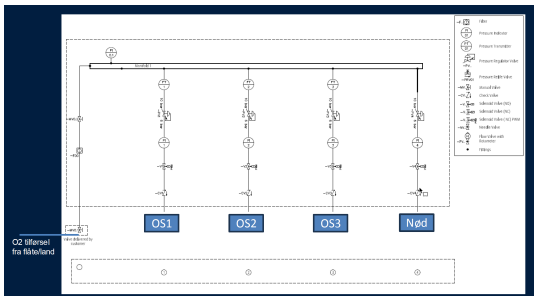
1. Inlet pump no. 1 is running at above minimum speed.
2. We have minimum pressure at oxygen piping manifold.
3. Operator sends start signal to the oxygenator system no. 1.

Scenarios



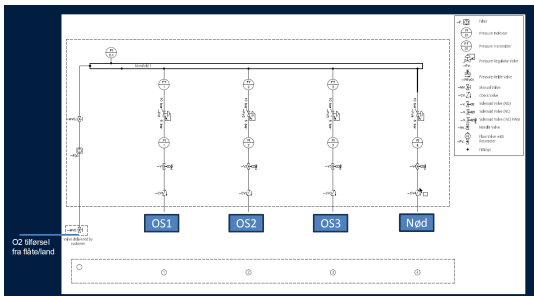
1. Inlet pump no. 1 is running at above minimum speed.
2. We have minimum pressure at oxygen piping manifold.
3. Operator sends start signal to the oxygenator system no. 1.
4. Control valve is set to a manual predefined position.

Scenarios



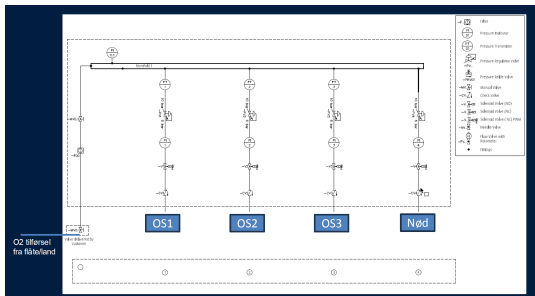
1. Inlet pump no. 1 is running at above minimum speed.
2. We have minimum pressure at oxygen piping manifold.
3. Operator sends start signal to the oxygenator system no. 1.
4. Control valve is set to a manual predefined position.
5. Solenoid valve opens.

Scenarios



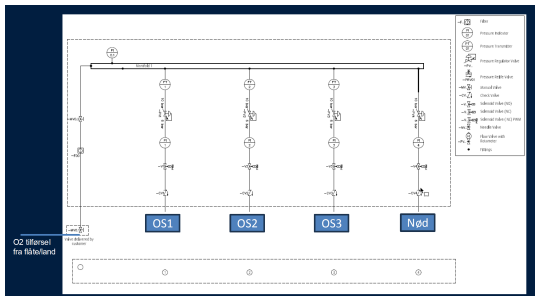
1. Inlet pump no. 1 is running at above minimum speed.
2. We have minimum pressure at oxygen piping manifold.
3. Operator sends start signal to the oxygenator system no. 1.
4. Control valve is set to a manual predefined position.
5. Solenoid valve opens.
6. Operator sends "auto mode" signal.

Scenarios



1. Inlet pump no. 1 is running at above minimum speed.
2. We have minimum pressure at oxygen piping manifold.
3. Operator sends start signal to the oxygenator system no. 1.
4. Control valve is set to a manual predefined position.
5. Solenoid valve opens.
6. Operator sends "auto mode" signal.
7. Operator changes set point to the desired oxygen concentration.

Scenarios



1. Inlet pump no. 1 is running at above minimum speed.
2. We have minimum pressure at oxygen piping manifold.
3. Operator sends start signal to the oxygenator system no. 1.
4. Control valve is set to a manual predefined position.
5. Solenoid valve opens.
6. Operator sends "auto mode" signal.
7. Operator changes set point to the desired oxygen concentration.
8. Operator verifies that the valve control signal and oxygen concentration levels are stable and moving towards a stationary value.